Benchmark Control Problem for Real-time Hybrid Simulation

-Companion Package-

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Contents

1 Getting started

1.1 Required software

- MATLAB R14 or later
- MATLAB Simulink Toolbox
- MATLAB Control Toolbox
- Simulink Desktop Real-Time[1]
    - Real-time Kernel[2,3]

1.2 Running the virtual RTHS demo

To execute the vRTHS demo follow the steps below.

1. Start MATLAB and change to the directory where the Benchmark files are located.
2. Run file `RUN_vRTHS_demo.m`. This file will sequentially execute the following scripts:
    (a) `F1_input_file.m`. This script will ask users if they want to run the simulation in real-time. This option requires a pre-installed real-time kernel that interacts with your operating system. (Please go to Section 3 for further details of how to install and execute the Simulink Desktop Real-time). Also, this script will request users for the number of additional runs. The first run is always the nominal plant (with actuator model's parameters set by the mean values described in the companion paper). The additional runs correspond to perturbed plants (with plant parameters randomly selected from normal distributions described in the companion paper). Users can also choose the building case to run from `case_1` to `case_4`. The case will be loaded from the `Building cases` folder. Aditionally, users can modify simulation and result parameters. For example, select the ground motion excitation, show the Bode plot of the controller, and show the RTHS system response.
    (b) `F2_controller.m`. This script loads the sample controller and compensator gains (see companion paper), verifies for stability, and generates (if desired) the transfer function plots of the controller.
    (c) `F3_simulation.m`. This script executes the Simulink model and plots the responses (if this option was selected).
    (d) `F4_evaluation.m`. This script calculates the evaluation criteria indices (described in the companion paper) and prints a table with the results.

After the execution of the simulation, the results are stored in the folder called `Results` for further analysis and post-processing. Additional files in the companion package are:

- Open file `Reference_def.m`, which is the code for generating the reference structure. In this file, users can define the mass and damping of the reference structure. Four building cases are already created and described in the companion paper. The cases will be saved in the in the `Building cases` folder.
- The MATLAB Live Editor `Controllers_design.mlx` file provides a step-by-step guide for designing and evaluating a PID tracking controller and a phase-lead compensator.

[1]Needed only if the benchmark code is executed in real-time
[2]Refer to section 3
[3]Install the Kernel Using MATLAB: https://www.mathworks.com/help/sldrt/ug/real-time-windows-target-kernel.html

1.3 Changing the Sample (default) Control Algorithm

To implement their controller, users need to modify file `F2_controller.m` and `controller_model.slx`. These files may be modified or even replaced by users to define their controllers in discrete state-space form. The Simulink file `controller_model.slx` contains the sample controller scheme and it is included in the main platform (`vRTHS_MDOF_SimRT2014a.slx`) as a *Referenced model* called *Controller*. This implementation allows to include a model as a block in another model. The Model block called *Controller* in the main platform displays input and output ports corresponding to the top-level input and output ports of the referenced model. Using these ports allow to connect any user control Simulink model as a referenced model in the Benchmark platform.

2 Companion Files on the Benchmark Problem Statement

The color key, shown in Table 1 indicates whether a file should be modified by users completely, partially or not at all. A brief description of each file is provided in Table 2. For an example MATLAB script portion, see Listing 1 where an editable part in file `F1_input_file.m` is shown.

Listing 1: Editable portion of the code

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%                     General Simulation Parameter                        %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%  NOTE: Participants should modify this portion of the file to choose
%        the simulation parameters.

eq_intensity = .4;           % EQ Intensity
E_sw = 1;                    % Ground motion switch:
                             %   1: El centro, 2: Kobe, 3: Morgan,  4: Chirp input
```

Table 1: Color key

Modify	Modify portions of	Do not modify

Table 2: List of files

Type	File	Description
m-files	Run_vRTHS_MDOF.m	This file runs the files of the virtual RTHS sequentially, computes and plots the results.
	F1_input_file.m	This file loads the reference model, defines the experimental substructure and estimates the numerical substructure thereof. It also defines the hydraulic actuator parameters and establishes the uncertainties for modeling the variability in some of them.
	F2_controller.m	This program loads the sample tracking controller for the Benchmark Problem. It also allows loading a phase compensator if-need-be. Participants should change specified portions of this file to suit the needs of their particular controller.
	F3_simulation.m	This file sets up the general simulation parameters. Participants should not change this file.
	F4_evaluation.m	This script calculates the evaluation criteria from the simulation outputs. Also checks that the actuator does not exceed its physical limits. It has been written to be used with the sample controllers. Designers should not modify this file. The output of `F4_evaluation` are files (Data_year_month_day_min_sec_(Nom or Pert)_#.mat) that contain the RTHS simulation results for post-processing.
	Reference_def.m	This file builds the FE model (30x30) and the reference model (3x3) and creates the reference state space matrices for four structural cases. The cases are stored in the Building cases folder.
mlx-file	Controllers_design.mlx	This MATLAB Live Editor file provides a step-by-step guide for designing and evaluating a PID tracking controller and a phase-lead compensator.
Functions	MM_mat.m	Function for calculating the mass matrix of the 12-DOF model. Also, the reduced order mass matrix for the 3-DOF reference model is built here after the rotational masses have been removed.
	KK_mat.m	Function for calculating the stiffness matrix of the 12-DOF model. Also, the reduced order stiffness matrix for the 3-DOF reference model is built after the rotational DOF have been condensed.
	CC_mat.m	Function for calculating the proportional damping matrix of the reference model.
	modes.m	Function for calculating the modal parameters of the reference model by solving the eigenvalue problem.
mat-files	plant.mat	This file contains the model of the plant to be controlled. The model is based on the the transfer system and the experimental substructure dynamics. This file is used in `Controllers_design.mlx` to design the tracking controller.
	ElCentroAccelNoScaling.mat	El Centro 1940 historic ground acceleration record without scaling.
	KobeAccelNoScaling.mat	Kobe 2008 historic ground acceleration record without scaling.
	MorganAccelNoScaling.mat	Morgan Hill 1984 historic ground acceleration record without scaling.
slx-files	vRTHS_MDOF_SimRT2014a.slx	This Simulink model is the simulation platform of the Benchmark Problem in RTHS. Users can replace the <control block> and modify the <sensors> block with their own to run their designs.
	controller_model.slx	Designers can replace this file with their own controller as a referenced model in a separate Simulink file. The controller design, in discrete form, should be here.

3 Simulink Desktop Real-Time

The Simulink Desktop Real-Time software requires a real-time kernel that interfaces with the operating system. Users may refere to `https://www.mathworks.com/help/sldrt/ug/real-time-windows-target-kernel.html` where detailed instructions on how to install Real-Time Kernel are provided. The following paragraph is a summary of the instruction provided by MathWorks.

The Simulink Desktop Real-Time kernel assigns the highest priority of execution to the users' real-time executable, which allows it to run without interference at the selected sample rate. During real-time execution, the kernel prioritizes CPU usage to execute each model update at the prescribed sample times. Once a model update completes, the kernel releases the CPU to run other operating system-based applications that need servicing. Note that the controller that users choose to implement must be able to execute in real time. This constraint should be evaluated using the virtual real-time option.

Install the Kernel Using MATLAB

Users must install the kernel before they can run a Simulink Desktop Real-Time application. During software installation, the Simulink Desktop Real-Time software is copied onto your hard drive, but the Simulink Desktop Real-Time kernel is not automatically installed into the operating system. Installing the kernel configures it to start running in the background each time users start their computer. **Failing to run a real-time simulation without having installed the kernel may cause a major crash of the computer due to insufficient computational resources that may lead to loss of data or unsaved work**. The following procedure describes how to use the command `sldrtkernel -install` . (users can also use the command sldrtkernel -setup instead.) To install the kernel:

1. Save all data and work and close all applications except MATLAB.

2. In the MATLAB Command Window, type:
   `sldrtkernel -install`
   The MATLAB Command Window displays one of these messages:
   users are going to install the Simulink Desktop Real-Time kernel.
   Do you want to proceed? [y] :

3. Type `y` to continue installing the kernel, or `n` to cancel without changing the current configuration.

4. After installing the kernel, check the installation by typing:
   `rtwho`
   If installed successfully, users should see the following message:
   `Simulink Desktop Real-Time version xxxx (C)`
   `The Mathworks, Inc. 1994-20xx`
   `Running on 64-bit computer`, (xxxx indicates users' system version and year)

Once the kernel is installed, it remains in idle mode. It's okay to leave it installed. While the kernel is idle, the operating system controls the execution of standard applications, such as Internet browsers, word processors, and the MATLAB environment. The kernel becomes active when users begin execution of their model, and becomes idle again after model execution completes.

What should users do if their computer crashes after installing the kernel?

There is a patch available from Mathworks that needs to be installed. This can be downloaded directly from the Mathworks website (`https://www.mathworks.com/support/bugreports/1719571`. However, the user needs to have a Mathworks account (which can be created for free). The user should carefully follow the instructions provided in the website. To install this patch, the user must also have administrative privileges for the terminal where MATLAB is installed.